

Sequential Variety in Work Processes

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Comments welcome, but please do not cite or circulate.
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Abstract

Task variety is a familiar concept in organization theory and in job analysis. Standardized scales to measure task variety have been developed and shown to be reliable, but results reported here suggest that while these scales may measure the number of different tasks performed, they do not measure variety in work *process*. Comparing results from the standard, survey-based measures and detailed, process-based measures in four task units, we discovered that work processes in the most “routinized” task units (as measured by the standard scales) are more varied than the least “routinized” task unit. The theoretical implications of this new concept, sequential variety, are discussed.

INTRODUCTION

Task variety is a familiar concept in organization theory (Perrow, 1967; Lynch, 1974; Daft and Macintosh, 1981) and in job analysis and design (Ilgen and Hollenbeck, 1992; Harvey, 1992). At the organizational level, task variety is seen as a property of the task unit -- a structurally defined sub-unit in an organization. At the individual level, task variety is seen as a property of a job and is a function of the number of different task elements the job requires (Hollenbeck and Ilgen, 1992). Since the early sixties (Hall, 1962), we have operationalized task variety at the organizational level by asking members of a task unit whether they agree with statements like, "People working here do about the same job in the same way most of the time" and "People working here perform repetitive activities in doing their jobs" (Withey, Daft and Cooper, 1983). At the job level, we use similar items to assess task variety, such as "The job is quite simple and repetitive" (Hackman and Oldham, 1980).

While these measures have considerable face validity and have been shown to be reasonably reliable (Withey, Daft and Cooper, 1983), evidence presented here suggests that they should not be interpreted as measures of variation in *work processes*. The emphasis here on process is important: while the existing literature offers operationalizations for task variety within jobs and task units, neither of these structurally defined units of analysis is necessarily related to work processes *per se*. Processes frequently span multiple jobs or task units, and examining these units in isolation may or may not reveal much about the workings of the organization. The question is, do self-report measures based on individual or task-unit level of analysis capture variability in work processes, or is variability in work processes an independent construct?

To investigate this question, we need to explicitly consider the sequential structure of work processes. In this paper, we use two methods for doing this: one based on optimal string matching (Sankoff and Kruskal, 1983; Abbott and Hryack, 1990; Abbott, 1995) and another based on Markov models of event sequences (Gottman and Roy, 1990). The string matching method directly compares each sequence with every other sequence, and measures the similarity among them. The Markov chain method compares the observed data to the "null model" that the observations are completely random.

Observations that are closer to the random model are more varied than observations that are far from random. These techniques provide an explicit, process-based alternative for measuring the variation in work processes in a way that has not previously been reported.

Using data collected at the Citibank United States Citibanking Center (USCC), we compare several alternative ways of measuring task variety. The analysis suggests that standard measures of sub-unit task variety (Lynch, 1974; Daft and Macintosh, 1981; Withey, Daft, and Cooper, 1983) and job variety (Hackman and Oldham, 1980), do not reflect variation in work processes as one might expect. Task units where work is reported to be relatively routine (low task variety) appear to have more variety in their work processes than task units where work is reported to be highly varied. This finding replicates a preliminary study by Pentland, Shabana, Soe, Ward and Roldan (1994), but with larger sample sizes and a more diverse set of measures.

In addition to a methodological contribution, this paper demonstrates the use of sequential variety as a way of measuring task unit variability. Sequential variety refers to the number of different ways that work processes are performed in order to accomplish the work. This process-based perspective provides a way to measure the variability of organizational processes that may span multiple individuals or task units. In a sense, it integrates levels of analysis by looking at work processes, rather than individuals or task units. Because work processes embody the means by which organizations transform inputs to outputs, process variety is more closely connected, theoretically, to issues such as flexibility, quality, technology, coordination and performance, than the conventional measures at the job or sub-unit level.

The paper begins with a brief review of the traditional perspectives on task variety. We then introduce a new process-based framework and methods for analyzing process variety. After demonstrating the research methods and presenting our findings, we discuss the implications of this approach for the analysis of flexibility and customization in both services and manufacturing. We also discuss the strengths and limitations of our methods and offer some suggestions for future research.

THEORY

There has always been some question about the appropriate level of analysis at which to operationalize variety in organizational work processes. For example, Hage and Aiken (1969, p. 367) note the tension between the individual and organizational level of analysis. Perrow (1967) explicitly identifies whole organizations as the unit of analysis for his framework for analyzing technology and structure. This approach was in keeping with Woodward (1965) and others who sought theories concerning whole organizations as the unit of analysis. But the empirical literature quickly moved to the sub-unit level of analysis, as in Van de Ven and Delbeq (1974, p. 183) who define a work unit as “the smallest formal grouping of individuals in an organization.” To compute their measure of task variety, Van de Ven and Delbeq (1974) aggregate individual responses to questions derived from Hall (1963) and Hage and Aiken (1967). Their data showed marked variations within a single organization, so the shift in theoretical emphasis to the sub-unit level made sense. Nonetheless, the conceptual and operational problem identified by Hage and Aiken (1969) remains: is it a property of organizational structure or of individual jobs? The answer we propose here is neither: for organization theorists, variety should be thought of as a property of processes, not jobs or structures. To make this argument, we need to take a closer look at the conceptual literature at both the sub-unit and job level.

Task Variety at the Sub-unit Level

Perrow’s (1967) framework for analyzing organizational technology has been highly influential in shaping the concept of variety (and routineness) at the organizational and sub-unit level. Perrow was explicitly concerned with the relationship between technology and organizational structure, and he sought ways to conceptualize technology that would allow this relationship to be studied and compared across a wide range of settings. For this reason, Perrow (1967, p. 195) used a rather abstract definition of technology: “the actions that an individual performs upon an object with or without the aid of tools or mechanical devices, in order to make some change in that object.” In this framework, variety was conceptualized as “the number of exceptional cases encountered in the work, that is, the degree to which stimuli are perceived as familiar or unfamiliar (p. 195-196). It is interesting to note that in subsequent studies (e.g., Van de Ven and Delbeq, 1974; Withey, Daft and Cooper, 1983), Perrow’s

original dimension (perceived number of exceptions) is simply replaced by the label “task variety.” This substitution was done with remarkably little explanation; as a field, we seem to have taken it for granted that “perceived exceptions” and “task variety” are the same construct. This dimension, along with the analyzability of exceptions, have been used to characterize the degree of routinization in a task unit (Perrow, 1967; Lynch, 1974; Daft and Macintosh, 1981; Withey, Daft, and Cooper, 1983). These variables can then be correlated to other features of the work unit or its environment, such technology, uncertainty, power or control. As long as the theoretical and practical interest was in work units *per se*, this framework was reasonable. But as the emphasis in theory and practice shifts to processes, we need to reexamine our operationalization of these basic ideas.

Task Variety at the Job Level

In research on job analysis and design, task variety is seen as a property of a job (Harvey, 1991; Hollenbeck and Ilgen, 1991). In particular, it is a function of the number of different task elements required in the performance of the job. These elements may be formally defined parts of the job or emergent activities that workers develop on their own. Fleishman and Quaintance (1984) offer an encyclopedic taxonomy of generic task elements which can be used to compare jobs between different organizational contexts. Other frameworks for job analysis use similar dictionaries of task elements (Harvey, 1992). For example, Gael (1983, p. 60) provides an “action verb list” for use in describing task elements. McCormick (1979) also provides a list of task elements for job analysis. The more elements, the more variety. There are also subjective measures of variety, such as Hackman and Oldham’s (1980) Job Diagnostic Survey. These measures are very similar in wording to those used at the task unit level.

This conception of task variety has great practical value, as well, because it maps directly onto the problem of selecting and training individuals to perform the jobs in question. As Hollenbeck and Ilgen (1991) note, this conception of task variety underlies virtually all approaches to synthetic validation in the area of personnel selection (Hollenbeck and Whitener, 1988; Mossholder and Arvey, 1984).

While the job level conception of task variety has significant practical value, it has two major shortcomings. First, even within the job, it fails to account for possible variations in task performance. Two jobs may encompass similar elements, but may allow (or require) very different degrees of variation in the performance of those elements. For example, many customer service interactions are highly scripted and constrained (Leidner, 1993). This is especially true in the case of out-going telemarketing calls, but even when customers initiate the interaction, they may be required to produce identification numbers, zip codes or other information before their requests can be processed. In principle (and in some organizations, in practice, as well), even quite routine tasks like taking orders can be accomplished in a more open-ended, conversational interaction. As long as all the necessary fields are filled in, no set script is required. Second, most work processes encompass more than a single job. This is also a shortcoming of the task-unit perspective. By focusing on individuals or structural sub-units, which are convenient from a measurement point of view, traditional measures of task variety tend to reproduce the “functional silos” that organizations have struggled for years to minimize or eliminate. To the extent that organizations have been successful in this effort, our methods are obsolete: they focus on units of analysis that are of questionable relevance to the phenomenon.

The underlying problem with these conceptions of task variety is that they do not address the sequential structure of work. Work processes are inherently composed of sequences of steps or activities. These sequences and the steps that compose them may vary between task units, or they may vary from time to time within a given task unit. By summarizing sequential work process in terms of indicators like "numbers of exceptions" or "analyzability of search," organization theorists have abstracted away from the core phenomena. Furthermore, as operationalized in most research, these concepts depend on subjective self-reports from task unit members. Inexpensive, survey-based measures have been developed that are reasonably reliable (Withey, Daft, and Cooper, 1983), but given the difficulties with the underlying conceptual structure it is not clear what these instruments are measuring.

Research has shown that ratings of task variety by job incumbents do not necessarily match objective measures. Incumbent ratings do not necessarily correlate with ratings by others (Hollenbeck

and Ilgen, 1992). Social cues can also be significant (O'Reilly and Caldwell, 1979). The issue here is not simply that objective and subjective measures of task variety may differ. As Hollenbeck and Ilgen (1992, p.181) note, one would expect that subjective self-report of task variety by incumbents would be influenced by the same kinds of factors that influence any such perception. The issue is that conceptually, the construct being measured does not reflect the variety of the work processes of the organization. This is true whether the unit of observation is an individual job or a task unit, because processes cut across these traditional structural boundaries.

There is good reason to expect that understanding and measuring process variety is important, both theoretically and practically. First, it relates directly to flexibility. An organization that can enact a wider variety of processes is, almost by definition, more flexible (Piore and Sabel, 1983; Pine, Victor and Boynton, 1993). In competitive environments where flexibility is important, process variation may be related to performance or even organizational viability. Second, process variation is widely seen as a major contributor (or barrier) to quality. The basic thrust of standards like ISO 9000, for example, is to control process variation. Third, on a more conceptual level, a more varied repertoire is generally seen as indicative of organizational learning (Huber, 1991). Finally, in cases where increased variation is desired (for example, in response to customer requests), it entails increased needs for information processing and coordination. In short, many of our core propositions concerning the linkages between technology, organization, and environment can be recast in process terms -- if we have a way to operationalize the underlying construct.

A Process-based Alternative

To response to this problem, we propose a framework that explicitly accounts for the sequential structure of work processes. Instead of focusing on the properties of a task unit or a job, we propose to examine the structure of the work process directly and in considerable detail. To do so, we use the concepts of lexical and sequential variety (Pentland et al, 1994).

Lexical Variety. In a given work situation, there are a limited number of actions that members of a task unit typically perform. These can be determined inductively, using ethnographic techniques

(Spradley, 1979; Pentland, 1992), or one can employ an established scheme such as that developed by Fleishman and Quaintance (1984) or McCormick (1979). In either case, the list of actions performed in the workplace forms a kind of lexicon -- these are the elements from which work processes are constructed. While the lexicon of a given process may change over time (for example, in response to changing technology), we can generally assume that it is quasi-static for any given iteration of a process. In other words, typical processes are usually performed more quickly than they can be changed.

Sequential Variety. Lexical variety does not incorporate any information about the structure of the work process. In particular, it fails to account for the many ways in which actions can be sequenced to create variations in work processes. Consider the difference between a sales situation where the interaction is highly scripted (Leidner, 1993) and a situation where the salesperson improvises in response to the customer. The lexicon would be similar, but the sequences would presumably differ. To capture this aspect of process variety, we ask how many different ways members of a given task unit can combine the moves available to them. That is, for a given lexicon of moves, is the sequence always the same, or can they vary it depending on the circumstance? We will refer to this dimension as sequential variety.

MEASURING SEQUENTIAL VARIETY: TWO APPROACHES

Conceptually, measuring sequential variety is straightforward: how many different ways is a process performed? It is not feasible to simply count alternatives, since actual process performances vary in a large number of ways. To quantify these variations, we employ two strategies: (1) direct comparison of the observed sequences to each other; and (2) comparison of the set of observations to null model (a random markov process).

Direct comparison of observations to each other

String matching is a technique that has been used extensively in molecular biology to compare protein sequences, such as DNA (Sankoff & Kruskal, 1983). It has also been applied to speech recognition and other topics where two sequences of unequal length must be compared. It has the important advantage that it makes no particular assumptions about the statistical properties of the data used as input; it simply computes a distance measure. Recently, the technique has been applied to the analysis of sequential data in the social sciences, such as the careers of musicians (Abbott & Hrycak, 1990), the formation of professions (Abbott, 1991), and the systems implementation process (Sabherwal & Robey, 1993). Abbott (1995) provides a useful review of social science applications.

Conceptually, string matching is quite simple. String matching programs compute the distance between two strings by counting up the number of operations needed to transform one string into the other. The operations typically include substituting one element for another, or inserting or deleting elements. Each operation has a cost, and the distance between the strings is the total cost. If the sequences were all identical, then the distance would be zero. If the sequences diverged from each other in a single element (e.g., "aaa", "aba"), then the distance would be equal to one.¹ The technique is called "optimal" string matching because it finds the lowest cost set of operations to accomplish the transformation, thus insuring that the computed distances are unique and well-behaved (e.g., they obey the triangle inequality: $d(A,B) + d(B,C) \geq d(A,C)$). Distances computed in this way are called Levenshtein distances (Sankoff & Kruskal, 1983).

Since our objective is to estimate the amount of variation in a set of sequences, we can compute the distance between each sequence and every other sequence and take the average:

¹ An important factor in the distance computation is the cost assigned to insertion, deletion, and substitution operations. In the computations reported here, all of these costs are set equal to one. We did not have a strong rationale for imposing a different cost structure, and the literature suggests that the outcomes of string matching procedures are relatively robust to perturbations in the cost structure (Abbott & Hrycak, 1990; Sabherwal & Robey, 1993).

$$\text{Average distance} = \frac{1}{n(n-1)/2} \sum_{i=1}^n \sum_{j=i}^n d(i, j), (i < j) \quad (1)$$

where n equals the number of observed sequences and d(i,j) equals the Levenshtein distance between each sequence divided by the length of the longest string in the pair (Abbott and Hryack, 1990).

Normalizing by the length of the strings makes the distance metric meaningful when dealing with processes of differing length. The factor n(n-1)/2 is simply the number of pairs in a set of n sequences.

For very routine processes, the sequences should be nearly identical and the average distance should be near zero. For more varied processes, the measure will increase.

Comparison to a null model

Another strategy for measuring sequential variety is to compare the set of observed sequences to a known standard: a completely random process. A uniform, random Markov process can be represented by a transition matrix where each element a_{ij} is equal. In other words, every event is equally likely no matter which event has just occurred. To apply this measure to a set of observed sequences, one needs to compute a transition matrix (Gottman and Roy, 1990). This is done by simply counting the number of times each code follows another in the sequences. The count for each pair is divided by the total number of pairs to get transition matrix for the observations. The Euclidian distance of the observed transition matrix from a matrix where all the elements are equal, can be computed as follows:

$$\text{Distance from null model} = \left(\sum_{i=1}^L \sum_{j=1}^L \left(\frac{a_{ij} - \bar{a}_{ij}}{N} \right)^2 \right)^{1/2} \quad (2)$$

where N is the total number of transitions, L is the size of the lexicon and $\bar{a}_{ij} = \frac{N}{L^2}$. This measure

reflects the inverse of variation; greater deviation from the uniform random model indicates a less varied process.

METHODOLOGY

Site Descriptions

The Investigations Unit of the U.S. Citibanking Center is responsible for resolving the inquiries and problems of banking customers that cannot be resolved by branch office personnel or the regular call center staff. The Investigations unit handles nearly 30,000 investigations in a typical month, on average. Investigations originate in the main call center and are assigned to an individual investigator who has considerable discretion in how to approach the work. As long as the conclusion is appropriately documented, the specific steps taken to resolve the problem can and do vary widely.

There six sub-units, four of which were included in this study. For the purposes of this paper, I will refer to them as A, B, C and D. Typical problems include missing or unrecognized items on a statement (e.g., “I made a deposit last month and it’s not showing up on my statement.”) Of these units, A is by far the highest volume, since it services problems that are generated by a popular consumer “bill payment” plan offered by the bank. Unit A also handles the smallest number of different kinds of problems (as shown in Table 1, below). Unit B handles a slightly larger variety of problems, but at much lower volume. Units C and D handle the largest number of different types of problems.

The goal of the investigations unit is to resolve problems as quickly and accurately as possible, with specific performance targets for different kinds of problems. To do so, they interact with other units of the Bank, including the data center(s), branch offices, microfiche and document retrieval, and an accounting unit that makes actual debits and credits to customer accounts. They also interact with external credit reporting services. These organizational structures enable and constrain the investigation process in very concrete ways (e.g., “I need to get a copy of the customers’ check from microfiche...”). This unit is supported by a wide range of information systems that give them access to information on customer accounts, branch office operations, and other aspects of the internal accounting system. There are also systems to track the status of the work and assist in document preparation (e.g., form letters to customers).

In the investigations unit, certain personnel are more or less dedicated to certain tasks. For example, each section has an individual whose sole responsibility is to verify that all the appropriate paperwork is in place and that the disposition seems reasonable. Other people have responsibility for a broader range of tasks. Thus, individual job variety is not a good indicator of the overall variability of work handled in any of the subunits. One has to trace problems through the system, which may include several different individuals or sub-units, to understand the sequence of actions in a given performance of the process.

Data Collection

Four kinds of data were collected from each subunit: a sample of “closed cases” from the case management database, an individual survey, a set of work sequences collected using a process tracing technique developed specifically for this study, and a set of very brief “exception surveys” that were attached to the work sequences. Thus, we have a broad array of objective and subjective measures of task variety for each of the four units. These data collection methods are summarized in Table 1 and discussed in the following sections.

Table 1: Data Collection Summary

	A	B	C	D	Total
Cases per month	16,875	681	4008	6126	27,690
Most common cases	Re-Credit Proof of Payment	Payment not posted Payment posted incorrectly	Deposit not posted Withdrawal unrecognized	Misc. Debit Unrecognized Deposit not posted	
Kinds of cases	9	10	16	25	
Individual surveys	43	18	30	28	119
Work sequences	169	182	162	205	718
Exception surveys	877	115	492	138	1622

Closed cases. A comprehensive sample of 27,690 closed cases was collected from one month's activity in the investigations unit (September 1999). This data provided the number of each kind of case handled by each of the sub-units. For example, in sub-unit C, the two most common investigations were "Deposit not posted" and "Withdrawal unrecognized." This provides a useful baseline on the volume and variety of work handled in each of the four sub-units, as reported in Table 1.

Individual surveys. A survey was administered to members of the investigation unit. All members of each task unit were invited to participate, and nearly every member of each task unit returned a survey. Task variety was measured using the scale from Withey, Daft, and Cooper (1983) ($\alpha = 0.87$). Job variety was measured using the "skill variety" dimension of the Job Diagnostic Survey (Hackman and Oldham, 1980) ($\alpha = 0.72$). The two scales correlated strongly ($r = 0.617$, $n=119$, $p<.001$).

Work sequences. A total of 718 work sequences were collected. A sample of cases was selected by the supervisor from each of the four groups. The sample was basically a "time slice" of cases that came into each sub-unit, chosen to minimize the impact on operations. While not technically a random sample, there is no reason to expect that the cases chosen were somehow unrepresentative.

Direct observation of work sequences in the investigations unit was not practical because cases often take more than a day to complete. Thus, a special "action sheet" was designed to allow investigators to record their own actions. Each person who handled the case added an "action sheet" to the case. Thus, for each case, there would be several action sheets stapled to the case. Each person recorded their own activity on the case, in their own words, writing one "action" on each line of the action sheet. This procedure was designed by the supervisors for each sub-unit, and was field tested on a small sample of cases. After some revisions, an average of 3.6 sheets per case, with an average of 7.6 actions per case (minimum length = 2 actions, maximum = 26 actions). While there is no question that this procedure does not result in perfect recording of actions, each set of action sheets was reviewed by the unit supervisor.

Coding was performed by entering all of the work sequences into a specially designed MS Access database. A total of 5148 actions were typed in, verbatim, from the action sheets. By sorting

(and re-sorting) events across all task units, we inductively derived a set of categories for the actions that could be used across all four task units. This procedure resulted in a set of 40 action codes and a common coding scheme with which to compare the four task units. For purposes of illustration, ten of the most frequent actions are shown in table 3.

Table 3: Typical actions in the work sequences

Code	Description
1	Case Assigned
4	Logged in
5	Logged out
7	Verified (QC)
10	Letter to customer
11	Note in account file (database)
21	Do debit/credit tickets
24	reviewed acct information
29	Request documents
33	Handoff/Forward case

Table 4 shows a set of ten observations, five from each of two different kinds of cases handled in sub-unit A. Notice that the pattern of actions differs for the two kinds of cases. Note that the sequences begin with “Log in” and end with “Verified (QC)” – this was a common pattern across all four subunits. The steps in between vary widely, as suggested in table 4.

Table 4 : Work processes represented as coded sequence data

Type	Sequence of actions
Re-issue	4,18,26,26,35,11,9,37,7
	4,18,26,26,35,11,9,37,7
	4,24,18,36,33,37,4,33,35,33,5,7
	4,4,7,10,37,4,33,35,33,5,7
	4,24,18,36,33,37,4,33,35,33,5,7
Proof of payment	4,13,11,9,33,5,7
	4,11,10,9,33,5,7
	4,13,11,10,9,5,7
	4,10, 9,33,5,7

4,13,11,9,5,7

The string matching procedure works directly on the coded sequences, as shown in Table 4. To compute the distance from the random, you need to summarize the sequences in a transition matrix (Gottman and Roy, 1990).

Exception surveys. In addition to recording their actions, members of the investigations unit were asked to record their perceptions of each case before they handed it off. The response rate on these items was approximately 62%. This provides a measure of the perceived exceptions that is connected to actual instances of work being performed. One item asked whether the work they had just completed was “exceptional” (versus “typical”). Another item (reverse coded) asked whether the work “fit standard procedures” (versus “not fitting standard procedures”). These items correlate well ($r = 0.49$, $n=1622$, $p<.001$) and were combined to make a scale ($\alpha = 0.65$)

ANALYSIS AND FINDINGS

Perceived task variety and exceptions

Table 5 summarizes the results of the survey based measures of task variety and exceptions at each task unit. The cells in Table 5 contain the average for each task unit, with the standard deviations in parentheses.

Table 5 Survey results by sub-unit

	N	A	B	C	D	p
Task Variety	112	5.46 (1.14)	5.55 (1.00)	4.84 (1.30)	4.83 (1.30)	0.045
Job Variety (JDS)	118	3.62 (1.27)	3.68 (1.11)	4.50 (1.43)	4.48 (1.26)	0.007
Exceptions scale	1622	1.07 (0.30)	1.17 (0.57)	2.11 (1.03)	1.84 (0.94)	0.000
		Lower Perceived Variety		Higher Perceived Variety		

One-way ANOVA indicates that the sub-units are significantly different on all three measures. Post-hoc contrasts using the Bonferroni method indicate that units A and B are similar, and units C and D are similar, but significantly different from units A and B. Thus, all three measures support the conclusion that task variety is perceived to be lower in units A and B and higher in units C and D. Note that common method variance is not a threat to the validity of this conclusion, since the exceptions scale was anchored on actual cases and collected on a different survey instrument from a different sub-set of task unit members.²

Sequential variety

The sequential variety measures were computed on the full set of work sequences for each sub-unit. Thus, a sample of 169 sequences was processed for sub-unit A. Table 6 shows the results for each measure, with standard deviation shown in parentheses. Standard deviations were estimated by a bootstrap procedure (N=500), described below. Note that when the distance from the null model is *smaller*, the variability in the sample is larger (more random). The work sequences from task units A and B are closer to the random model than the sequences from units C and D. The string matching distance represents the average distance between each pair of sequences. For Unit A, for example, 169 sequences results in 14,196 pair-wise comparisons. A larger average distance implies more variation in the sequences. Thus, by this measure, the sequences from units A and B are more varied than the sequences from units C and D.

Table 6: Sequential variety results by sub-unit

	A	B	C	D
N	169	182	162	205
Distance from random model	0.148 (.0051)	0.191 (.0053)	0.225 (.0076)	0.222 (.0075)
String Match distance	0.259 (.0177)	0.183 (.0054)	0.143 (.0050)	0.110 (.0048)

² The correspondence of the standard task variety measures with the exceptions scale provides some support for the widespread re-interpretation of this dimension of Perrow's (1967) original framework.

	Higher variety	Lower Variety
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A bootstrap procedure was used to estimate the standard deviation for each of these parameters (Efron and Tibshirani, 1993), which allows us to make inferences concerning the statistical significance of the differences between the sub-units. In this procedure, one re-computes the test statistics on a “bootstrap sample”, drawn randomly, with replacement, from the set of observations. For example, for sub-unit A, 500 random samples of 169 sequences were drawn, with replacement, from the set of 169 actual observations. The standard deviations reported in table 6 are based on these 500 bootstrap samples.³ The contrast between C and D is not significant, but all of the others are at least two (and in most cases four or more) standard deviations away from each other ($p < .05$ or smaller). Thus, we can say that sub-units C and D are similar, but the others are significantly different on both measures of sequential variety. If we rank the four sub-units, we find that units A and B have relatively high sequential variety, while units C and D have relatively low sequential variety.

DISCUSSION

The data reported here raise some interesting questions concerning the interpretation of traditional measures of task variety. Table 7 summarizes the findings for all four sites by ranking each site “high” or “low” on each of the measures reported in this paper. The number of sub-units included in this study is small ($n=4$), but the pattern of findings suggests that traditional measures of task variety do not reflect sequential variety. The use of three different survey measures, and two different process-based measures lends support to this conclusion.

Table 7: Summary of findings

A B C D

³ Efron and Tibshirani (1993, p.13) note that 50-200 bootstrap samples are usually sufficient for estimation of the standard error.

Kinds of cases		9	10	16	25
Survey based measures	Task Var	L	L	H	H
	JDS	L	L	H	H
	Exceptions	L	L	H	H
Process based measures	Markov	H	H	L	L
	String Match	H	H	L	L

One proposition that can be drawn from this summary table is that traditional measures of task variety and job variety have more to do with the content of work than with the process of the work. Note that in terms of rank order, the two “high variety” units handle a larger number of different kinds of cases (e.g., “unrecognized withdrawal”, “missing deposit”, etc.). The two “low variety” units handle fewer kinds of cases. It is important to note that these categories are derived from a typical month of cases actually processed. Thus, it reflects the typical range of problems each unit handles. Since one part of the investigators job is to classify each case they handle, these categories directly reflect the participants own schema for the variety of problems they handle and the distinctions among them.

At the same time, while unit A handles only 9 kinds of cases, it does so with a wider variety of sequential processes than unit D, which handles 25 kinds of cases. This finding replicates that of Pentland et al (1994), who reported that reference librarians perceived their work to be highly varied compared to travel agents. This perception seemed to be based on the content of the work: reference librarians field questions on everything from medieval French poetry to recombinant DNA, while travel agents basically handle three things: air, hotel, rental car. Yet observational data suggested that the reference librarians actual work processes were much less varied than the travel agents. Thus, the variety in the content was opposite to the variety in the work process.

This suggests that when people are asked questions about their work that seemingly refer to the work process (e.g., “Do people working here do about the same job in the same way most of the time?”), they tend to respond in terms of the *content* of what they do, not the *process* of how they do it. While it is beyond the scope of this study to investigate the reasons for this, there are some plausible explanations based on differences in the way humans perceive and encode events versus objects.

First of all, people appear to have more detailed and reliable categorization schemes for objects (nouns) than for actions (verbs). Morris and Murphy (1990) suggest that event categorization is intrinsically less stable than object categorization. This is because actions have a number of components that can be separated in different actions. An action might include an actor, an object, a recipient and an activity, done in a certain manner in a certain location or context. However, at other times, the same activity might be done with a different actor, in a different manner, and so on. Different verbs encode different components of these actions, which suggests that there is no single, preferred way to conceptualize every action (see also Malone et al, 1999). The parts of objects, on the other hand, are often not physically separable from each other and do not occur alone. In addition, the boundaries of objects are much more clearly defined than those of actions. It is often very difficult to delineate precisely when an event has begun and ended. Morris and Murphy (1990), further, pointed out an additional factor that seems to explain the differences in stability of object and event categories. For some kind of objects, physical laws impose a categorical structure; whereas social events, and event categories, are shaped by human goals which are plastic and multiplex; as a result they are less stable.

Another reason to expect that perceptions of task variety reflect content more than process comes from studies on memory. For at least the last 70 years, the suggestion that there are different kinds of learning or distinct memory systems has received considerable support (Eichenbaum, 1997). Knowledge about things is stored in declarative memory and therefore easily subject to recall, while actions are stored in procedural memory and not so easily recalled (Schacter, 1989; Cohen and Bacdayan, 1994). Thus, the most salient aspects of the job environment are likely to be those aspects stored in declarative memory, such as the kinds of cases being handled. The sequences of tasks required to do the work is more likely to be part of procedural memory. Without some kind of process

tracing method to make the process visible, participants are less likely to be aware of just how varied (or routine) their work actually is.

Implications

One important theoretical implication of this research concerns our understanding of flexibility and adaptation. Increased flexibility within a task unit should also imply greater lexical and sequential variety. Adaptation should show up as changes in the lexicon or in the set of sequences produced (Pentland, 1995). We need to be mindful, however, of the distinction between process and content. Flexibility (and the closely related notion of adaptability) are intuitively connected to the content of work: how many different kinds of products or services can be provided, for example. The question we need to ask, we would argue, is how this is related to flexibility of work processes. It is quite possible that work processes may vary widely, appearing to be highly flexible without any meaningful ability to produce different kinds of goods or services.

There are a range of interesting questions that one can pose about lexical and sequential variety in task units. By now, the reader should have some intuition for what these constructs mean and how they might relate to more familiar ideas from organizational behavior. For example, we would expect that higher division of labor and increasing specialization would lead to sequential variety in each job description. As jobs become narrower, the actions required to accomplish them and the range of ways in which they can be accomplished should also be reduced. Conversely, we might expect that jobs that had been "reskilled" or "empowered" might exhibit a larger sequential variety than jobs that had not. In this usage, however, we have shifted the unit of analysis from the task unit to the specific job.

Some Methodological Issues

The concepts of lexical and sequential variety pose a number of interesting methodological challenges. The most important, perhaps, is how to get reliable data cheaply. The most reliable results can be obtained using video-taped data, but this approach is expensive and would seem to limit the

possible applications of the concepts to a rather narrow range of settings where videotaping is possible. Alternatively, one can engage in direct observations, but this approach makes it difficult to achieve high reliability (Folger, Hewes & Poole, 1984). As work processes increasingly become supported by work flow automation, of course, there will be an increasing possibility of using electronically generated data. Such an approach has obvious limitations, because the data would be filtered through a system that may or may not operationalize important theoretical constructs. Further, only a limited selection of sites would make such data available.

Also, these measures are strongly dependent on the point of view of the observer. Researcher judgments such as the granularity of the coding scheme and the boundaries on the process have a significant effect on the numerical values computed. In the current study, these factors were controlled for by the use of a common coding scheme and the process boundary across all four sub-units. Subsequent applications of these methods, if any, require the use of a uniform observational scheme or else the results are not meaningful.

Practical implications

The framework proposed here has implications for some areas of managerial practice, such as the design of automated systems to support work. While it is a truism that informal work practices rarely conform to official guidelines, there has been little attention to the issue of how to characterize these divergences. The notion of sequential variety starts to provide rigorous methodological handle on this issue. It can also provide some insight into what range of activities is actually being performed in a work unit. This kind of data is valuable input for a systems analysis and design.

Similarly, sequential variety provides a very concrete descriptive base from which to think about redesigning existing processes. Most process descriptions tend to be static -- as though work was accomplished the same way all the time. The concept of sequential variety directly challenges this view by calling attention to the innumerable ways in which work flow varies from instance to instance, even within supposedly highly routinized task units. Data of this kind may help alert process designers to potential problems in an existing process and signal a potential need for flexibility in the new process.

All processes have variety, and the drive to be increasingly responsive to customer needs seems to encourage still more variety and customization. The practical question is not how to limit variety through increased controls, but how to design systems that foster “good” variety (that results in increased customer satisfaction, for example) and limit “bad” variety (that results in substandard performance).

CONCLUSION

In this paper, we have proposed a new, process-based framework for thinking about task variety that builds on a linguistic metaphor for describing organizational processes. We introduced the concepts of lexical and sequential variety and we have begun to explore their relationship to traditional concepts of task variety and analyzability. The question, in a sense, is how to describe the routines and routineness in task units in a way that is theoretically valid and practically useful. We believe that the results reported here make some tentative steps in that direction.

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